21-1a Representing Consumption Opportunities in a Graph

To keep things simple, we examine the decision facing a consumer who buys only two goods: pizza and Pepsi. Although real people buy hundreds of different kinds of goods, assuming there are only two goods simplifies the problem without altering the basic insights about consumer choice.

We first consider how the consumer's income constrains the amount she spends on pizza and Pepsi. Suppose the consumer has an income of \$1,000 per month and spends her entire income on pizza and Pepsi. The price of a pizza is \$10, and the price of a liter of Pepsi is \$2.

The table in Figure 1 shows some of the many combinations of pizza and Pepsi that the consumer can buy. The first row in the table shows that if the consumer spends all her income on pizza, she can eat 100 pizzas during the month, but she would not be able to buy any Pepsi at all. The second row shows another possible consumption bundle: 90 pizzas and 50 liters of Pepsi. And so on. Each consumption bundle in the table costs exactly \$1,000.

Figure 1

The Consumer's Budget Constraint

The budget constraint shows the various bundles of goods that the consumer can buy for a given income. Here the consumer buys bundles of pizza and Pepsi. The table and graph show what the consumer can afford if her income is \$1,000, the price of pizza is \$10, and the price of Pepsi is \$2.

The graph in Figure 1 illustrates the consumption bundles that the consumer can choose. The vertical axis measures the number of liters of Pepsi, and the horizontal axis measures the number of pizzas. Three points are marked on this figure. At point A, the consumer buys no Pepsi and consumes 100 pizzas. At point B, the consumer buys no pizza and consumes 500 liters of Pepsi. At point C, the consumer buys 50 pizzas and 250 liters of Pepsi. Point C, which is exactly at the middle of the line from A to B, is the point at which the consumer spends an equal amount ($\$500$) on pizza and Pepsi. These are only three of the many combinations of pizza and Pepsi that the consumer can choose. All the points on the line from A to B are possible. This line, called the **budget constraint** (the limit on the consumption bundles that a consumer can afford) , shows the consumption bundles that a consumer can afford. In this case, it shows the trade-off between pizza and Pepsi that the consumer faces.

The slope of the budget constraint measures the rate at which the consumer can trade one good for the other. Recall that the slope between two points is calculated as the change in the vertical distance divided by the change in the horizontal distance ("rise over run"). From point A to point B, the vertical distance is 500 liters, and the horizontal distance is 100 pizzas. Thus, the slope is 5 liters per pizza. (Actually, because the budget constraint slopes downward, the slope is a negative number. But for our purposes we can ignore the minus sign.)

Notice that the slope of the budget constraint equals the *relative price* of the two goods—the price of one good compared to the price of the other. A pizza costs five times as much as a liter of Pepsi, so the opportunity cost of a pizza is 5 liters of Pepsi. The budget constraint's slope of 5 reflects the trade-off the market is offering the consumer: 1 pizza for 5 liters of Pepsi.